

DISSERTATION

TODO: DISSERTATION TITLE

Submitted by

José Ignacio Rojas Rojas

Department of Mathematics

In partial fulfillment of the requirements

For the Degree of Doctor of Philosophy

Colorado State University

Fort Collins, Colorado

TODO: Semester and year

Doctoral Committee:

Advisor: Renzo Cavalieri

Mark Shoemaker

Maria Gillespie

Joshua Berger

TODO: Copyright statement and year.

ABSTRACT

TODO: DISSERTATION TITLE

TODO: Write the dissertation abstract.

TABLE OF CONTENTS

ABSTRACT	ii
Chapter 1 Introduction	1
Chapter 2 Background	2
2.1 Category Theory	2
2.1.1 Examples of categories	3
Chapter 3 Pseudostable Maps and the Main Construction	5
Bibliography	6

Chapter 1

Introduction

Chapter 2

Background

2.1 Category Theory

Definition 2.1. A category C consists of

- A class Obj of entities called objects.
- A class Arw of entities called arrows.
- Assignments

$$source : Arw \rightarrow Obj, \quad \text{and} \quad target : Arw \rightarrow Obj.$$

These assignments are represented in the obvious way: $A \xrightarrow{f} B$ indicates that f is an arrow with *source* A and *target* B .

- An assignment

$$1 : Obj \rightarrow Arw$$

which, given an object $A \in C$, yields an arrow 1_A satisfying $A \xrightarrow{1_A} A$. In other words, the *source* and *target* assignments of the distinguished arrow 1_A are A itself.

- A partial composition

$$Arw \times Arw \rightarrow Arw.$$

Given two arrows $A \xrightarrow{f} B_1$ and $B_2 \xrightarrow{g} C$, they are composable when B_1 and B_2 are the same object. The resulting arrow has the form $A \xrightarrow{fg} C$.

Additionally the category C must satisfy the following conditions:

- Suppose we are given a diagram as follows:

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D.$$

We require that $(fg)h = f(gh)$. In other words, composition is associative.

- Consider an arrow f and the two identity arrows as follows:

$$A \xrightarrow{1_A} A \xrightarrow{f} B \xrightarrow{1_B} B.$$

We require that $f1_B = f = 1_A f$.

Remark 2.2. We say “class” and “assignment” to allow for the possibility that Obj and Arw are not sets. If they were, then an assignment would be the same as a function. We also denote, $Mor_C(A, B)$ for the class of all arrows with source A and target B .

2.1.1 Examples of categories

Example 2.3. In these categories, every object has extra properties and arrows are functions which respect this property. Rather than making the notion of *concrete category* precise, we will list some examples:

Category	Objects	Arrows
Set	sets	functions
Grp	groups	morphisms
AbGrp	Abelian groups	morphisms
Rng	rings	morphisms
Top	topological spaces	continuous functions
Vect _k	vector spaces over a field k	linear maps

Definition 2.4. For a category C , D is a subcategory of C if the class of objects (resp. arrows) of D is a subset of the class of objects (resp. arrows) of C .

Moreover D is a full subcategory of C if it's a subcategory and for all objects X, Y of D ,

$$Mor_D(X, Y) = Mor_C(X, Y).$$

An example of an important category that will be central in our study is the following:

Example 2.5. The category of schemes Sch has as its **objects** ringed spaces $(X, \mathcal{O}_X)$, where $\mathcal{O}_X$ is a sheaf of rings on a topological space X , such that any point of X has an open neighborhood U with the property that $(U, \mathcal{O}_X|_U)$ is an affine scheme.

Consider now an **arrow** $(X, \mathcal{O}_X) \xrightarrow{f} (Y, \mathcal{O}_Y)$.

Chapter 3

Pseudostable Maps and the Main Construction

Bibliography

- [1] Dave Anderson and William Fulton. *Equivariant Cohomology in Algebraic Geometry*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2025.
- [2] Enrico Arbarello and Maurizio Cornalba. Calculating cohomology groups of moduli spaces of curves via algebraic geometry, 1998.
- [3] Enrico Arbarello and David Mumford. *The Red Book of Varieties and Schemes: Includes the Michigan Lectures (1974) on Curves and their Jacobians*. Lecture Notes in Mathematics. Springer Berlin Heidelberg, 1999.
- [4] Luca Battistell, Francesca Carocci, and Cristina Manolache. Virtual classes for the working mathematician. *Symmetry, Integrability and Geometry: Methods and Applications*, April 2020.
- [5] Luca Battistella, Francesca Carocci, and Cristina Manolache. Reduced invariants from cuspidal maps, 2018.
- [6] Kai Behrend. Gromov–witten invariants in algebraic geometry. *Inventiones Mathematicae*, 127(3):601–617, 1997.
- [7] Kai Behrend and Barbara Fantechi. The intrinsic normal cone. *Inventiones Mathematicae*, 128(1):45–88, 1997.
- [8] Renzo Cavalieri, Joel Gallegos, Dustin Ross, Brandon Van Over, and Jonathan Wise. Pseudostable hodge integrals, 2022.
- [9] Renzo Cavalieri, Maria Gillespie, and Leonid Monin. Projective embeddings of $\overline{M}_{0,n}$ and parking functions, 2021.
- [10] Renzo Cavalieri and Matthew M. Williams. Quadratic pseudostable hodge integrals and mumford’s relations, 2024.

- [11] Carel Faber and Rahul Pandharipande. Hodge integrals and gromov-witten theory, 1998.
- [12] William Fulton. *Intersection Theory*. Mathematics and Statistics. Springer New York, 2012.
- [13] William Fulton and Rahul Pandharipande. Notes on stable maps and quantum cohomology, 1997.
- [14] Ezra Getzler and Rahul Pandharipande. Virasoro constraints and the chern classes of the hodge bundle. *Nuclear Physics B*, 530(3):701–714, October 1998.
- [15] Oliver Goertsches and Leopold Zoller. Equivariant de rham cohomology: Theory and applications, 2019.
- [16] Tom Graber and Rahul Pandharipande. Localization of virtual classes. *Inventiones Mathematicae*, 135(2):487–518, 1999.
- [17] Tom Graber and Rahul Pandharipande. Constructions of nontautological classes on moduli spaces of curves, 2003.
- [18] K. Hori, S. Katz, A. Klemm, R. Pandharipande, R. Thomas, C. Vafa, R. Vakil, and E. Zaslow. *Mirror Symmetry*. Clay Mathematics Monographs. American Mathematical Society, 2023.
- [19] Mikhail M. Kapranov. Veronese curves and Grothendieck-Knudsen moduli space $\overline{M}_{0,n}$. *J. Algebraic Geom.*, 2(2):239–262, 1993.
- [20] Joachim Kock. Notes on psi-classes. <http://mat.uab.cat/~kock/GW/notes/psi-notes.pdf>, 2001.
- [21] Joachim Kock and Israel Vainsencher. *An Invitation to Quantum Cohomology: Kontsevich’s Formula for Rational Plane Curves*. Progress in Mathematics. Birkhäuser Boston, 2006.
- [22] Jeongseok Oh. An introduction to virtual cycles—via classical algebraic geometry. In Young-Hoon Kiem, editor, *Moduli Spaces, Virtual Invariants and Shifted Symplectic Structures*, pages 65–94. Springer Nature Singapore, Singapore, 2025.
- [23] Dustin Ross. Pseudo-stable localization. unpublished, 2013.

- [24] Johannes Schmitt and Jason van Zelm. Intersections of loci of admissible covers with tautological classes, 2018.
- [25] David Schubert. A new compactification of the moduli space of curves. *Compositio Mathematica*, 78(3):297–313, 1991.
- [26] Julianna S. Tymoczko. An introduction to equivariant cohomology and homology, following goresky, kottwitz, and macpherson, 2005.
- [27] Ravi Vakil. *The Rising Sea: Foundations of Algebraic Geometry*. Princeton University Press, 2025.
- [28] Michael Viscardi. Alternate compactifications of the moduli space of genus one maps, 2010.
- [29] Stephanie Yang. Calculating intersection numbers on moduli spaces of pointed curves, 2010.
- [30] Dimitri Zvonkine. An introduction to moduli spaces of curves and their intersection theory. <https://www.hse.ru/data/2014/11/13/1101420183/2013.04.03.pdf>, 2014.